PRACTICE SET – UNIT I

Q1. Find the rank of the following matrices by reducing them to echelon form.

(1) $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$ **Ans.:** $\text{rank}(A) = 2$

(2) $A = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 2 \\ 2 & 1 & -3 \end{bmatrix}$ **Ans.:** $\text{rank}(A) = 3$

(3) $A = \begin{bmatrix} 0 & 1 & 2 & -2 \\ 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 1 \end{bmatrix}$ **Ans.:** $\text{rank}(A) = 2$

(4) $A = \begin{bmatrix} 2 & 4 & 3 & -2 \\ -3 & -2 & -1 & 4 \\ 6 & -1 & 7 & 2 \end{bmatrix}$ **Ans.:** $\text{rank}(A) = 3$

(5) $A = \begin{bmatrix} 3 & 4 & 1 & 1 \\ 2 & 4 & 3 & 6 \\ -1 & -2 & 6 & 4 \\ 1 & -1 & 2 & -3 \end{bmatrix}$ **Ans.:** $\text{rank}(A) = 4$

(6) $A = \begin{bmatrix} 1 & 4 & 3 & -2 & 1 \\ -2 & -3 & -1 & 4 & 3 \\ -1 & 6 & 7 & 2 & 9 \\ -3 & 3 & 6 & 6 & 12 \end{bmatrix}$ **Ans.:** $\text{rank}(A) = 2$

Q2. Find the rank of the following matrices by reducing them to normal form.

(1) $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 2 & 3 \end{bmatrix}$ **Ans.:** $\text{rank}(A) = 3$

(2) $A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 3 & 1 \\ 6 & 8 & 7 & 5 \end{bmatrix}$ **Ans.:** $\text{rank}(A) = 4$

(3) $A = \begin{bmatrix} 3 & 2 & 5 & 7 & 12 \\ 1 & 1 & 2 & 3 & 5 \\ 3 & 3 & 6 & 9 & 15 \end{bmatrix}$ **Ans.:** $\text{rank}(A) = 2$

(4) $A = \begin{bmatrix} 2 & -4 & 3 & 1 & 0 \\ 1 & -2 & 1 & -4 & 2 \\ 0 & 1 & -1 & 3 & 1 \\ 4 & -7 & 4 & -4 & 5 \end{bmatrix}$ **Ans.:** $\text{rank}(A) = 3$

Q3. Test the consistency of the following systems of equations. If consistent, solve them.

(i) **Ans.: Inconsistent.**

$$\begin{aligned} 2x - 3y - z &= 3, \\ x + 2y - z &= 4, \\ 5x - 4y - 3z &= -2. \end{aligned}$$

(ii) **Ans.: Consistent;** $x = 2,\ y = 1,\ z = 3$.

$$\begin{aligned} x + 2y - z &= 1, \\ x + y + 2z &= 9, \\ 2x + y - z &= 2. \end{aligned}$$

(iii) **Ans.: Consistent;** $x = -1,\ y = -2,\ z = 4$.

$$\begin{aligned} 6x + y + z &= -4, \\ 2x - 3y - z &= 0, \\ -x - 7y - 2z &= 7. \end{aligned}$$

(iv) **Ans.: Consistent; infinitely many solutions:**

$$\begin{aligned} 2x - y - z &= 2, \\ x + 2y + z &= 2, \\ 4x - 7y - 5z &= 2. \end{aligned}$$

$$x = \frac{6+t}{5}, \quad y = \frac{2-3t}{5}, \quad z = t.$$

(v) **Ans.: Consistent; infinitely many solutions:**

$$\begin{aligned} 2x_1 + x_2 + 5x_4 &= 4, \\ 3x_1 - 2x_2 + 2x_3 &= 2, \\ 5x_1 - 8x_2 - 4x_3 &= 1. \end{aligned}$$

$$\begin{aligned} x_1 &= \frac{53}{35} - \frac{12}{7}t, \\ x_2 &= \frac{34}{35} - \frac{11}{7}t, \\ x_3 &= t - \frac{3}{10}, \qquad x_4 = t. \end{aligned}$$

Q4. Determine the values of the parameters for which the following systems have a unique solution, infinitely many solutions, or no solution.

(a)

$$\begin{aligned} x + 2y + z &= 3, \\ x + y + z &= \lambda, \\ 3x + y + 3z &= \lambda^2. \end{aligned}$$

Ans.: Unique: None; Infinitely many: $\lambda = 2, 3$**; No solution:** $\lambda \neq 2, 3$**.**

(b)

$$\begin{aligned} x + y + z &= 1, \\ 2x + y + 4z &= k, \\ 4x + y + 10z &= k^2. \end{aligned}$$

Ans.: Unique: None; Infinitely many: $k = 1, 2$**; No solution:** $k \neq 1, 2$**.**

(c)

$$\begin{aligned} x + y + \lambda &= 6, \\ x + 2y + 3z &= 10, \\ x + 2y + \lambda z &= \mu. \end{aligned}$$

Ans.: Unique: $\lambda \neq 3$**,** μ **arbitrary; Infinitely many:** $\lambda = 3,\ \mu = 10$**; No solution:** $\lambda = 3,\ \mu \neq 10$**.**

(d)

$$\begin{aligned} 2x - 3y + 6z - 5t &= 3, \\ y - 4z + t &= 1, \\ 4x - 5y + 8z - 9t &= \lambda. \end{aligned}$$

Ans.: Unique: None; Infinitely many: $\lambda = 7$**; No solution:** $\lambda \neq 7$**.**

Q5. The electrical circuit shown in the figure consists of resistors and voltage sources. Using Kirchhoff's laws, formulate the system of simultaneous linear equations and find I_1**,** I_2**, and** I_3**.**

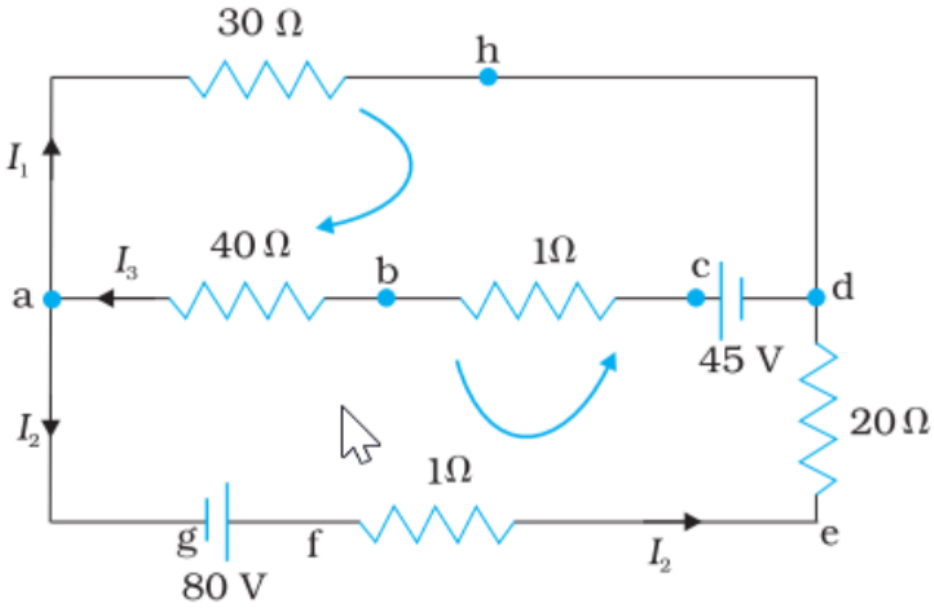


Figure 1:

Ans: $I_1 = -\dfrac{2335}{2721}$ A, $\quad I_2 = \dfrac{7030}{2721}$ A, $\quad I_3 = \dfrac{1565}{907}$ A.

Q.6 For the electrical circuit shown in the figure, use Kirchhoff's laws to formulate the system of simultaneous linear equations corresponding to the circuit. Hence, determine the currents I_1**,** I_2**, and** I_g**.**

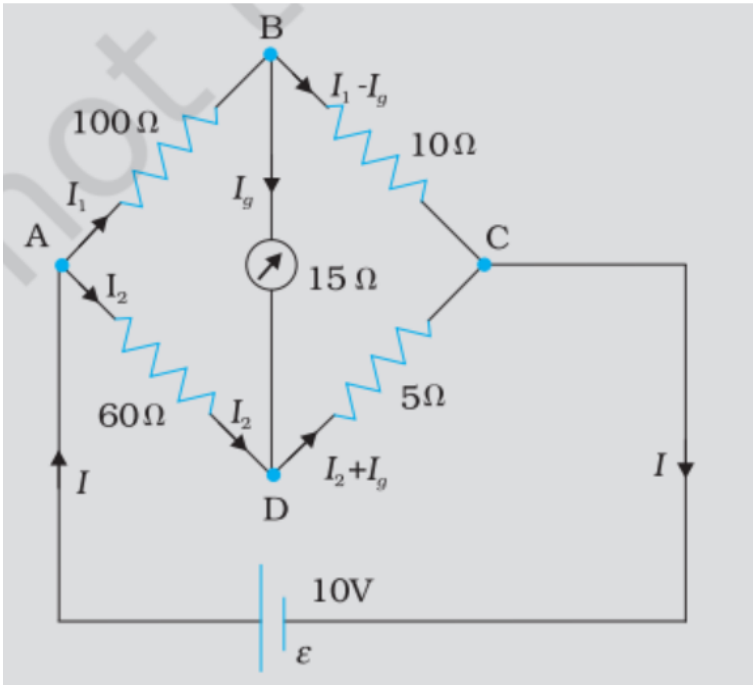


Figure 2:

Ans: $I_1 = 0.091$ A, $I_2 = 0.153$ A, $I_g = 0.005$ A.

Q.7 For the electrical circuit shown in the figure, use Kirchhoff's laws to formulate the system of simultaneous linear equations corresponding to the circuit. Hence, determine the currents I_1, I_2, and I_g.

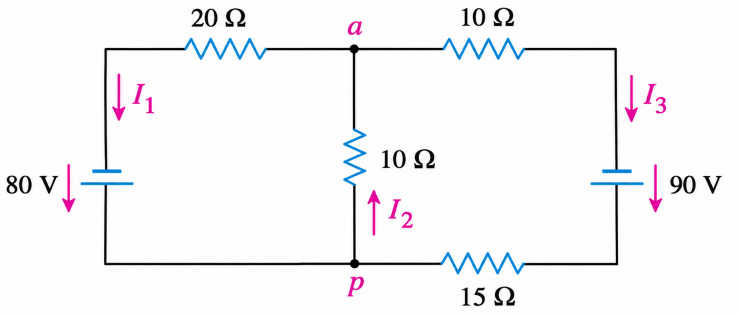


Figure 3:

Ans: $I_1 = 2$ A, $I_2 = 4$ A, $I_3 = 2$ A.

PRACTICE SET – UNIT II

Q1. Examine the following systems of vectors for linear dependence. If dependent, find the relation between them.

(1)

$$X_1 = (-1, -1, 1), \quad X_2 = (2, 1, 1), \quad X_3 = (3, 0, 2).$$

Ans.: Independent.

(2)

$$X_1 = (1, 2, 3), \quad X_2 = (2, -2, 6).$$

Ans.: Independent.

(3)

$$X_1 = (3, 1, -4), \quad X_2 = (2, 2, -3), \quad X_3 = (0, -4, 1).$$

Ans.: $2X_1 - 3X_2 - X_3 = 0$; Dependent.

(4) Find whether the following sets of vectors are linearly dependent or independent:

(i) $(1, -2), (2, 1), (3, 2)$,

(ii) $(1, 1, 1, 1), (0, 1, 1, 1), (0, 0, 1, 1), (0, 0, 0, 1)$.

Ans.: (i) Dependent (ii) Independent.

Q2. Find the eigenvalues and the corresponding eigenvectors for the following matrices.

(1)

$$A = \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}.$$

Ans.: $\lambda = -1, 1, 2$; eigenvectors: $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$.

(2)

$$A = \begin{bmatrix} 4 & 2 & -2 \\ -5 & 3 & 2 \\ -2 & 4 & 1 \end{bmatrix}.$$

Ans.: $\lambda = 1, 2, 5$; eigenvectors: $\begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.

(3)

$$A = \begin{bmatrix} -9 & 2 & 6 \\ 5 & 0 & -3 \\ -16 & 4 & 11 \end{bmatrix}.$$

Ans.: $\lambda = -1, 1, 2$; eigenvectors: $\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$.

(4)

$$A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}.$$

Ans.: $\lambda = -1, 1, 4$; eigenvectors: $\begin{bmatrix} -6 \\ -2 \\ 7 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}$.

Q3. Verify whether the following matrices are orthogonal. If orthogonal, write A^{-1}.

(1) If

$$A = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & a \\ \frac{2}{3} & \frac{1}{3} & b \\ \frac{2}{3} & -\frac{2}{3} & c \end{bmatrix}$$

is orthogonal, find a, b, c. **Ans.:** $a = \pm\frac{2}{3},\ b = \mp\frac{2}{3},\ c = \pm\frac{1}{3}$.

(2) Find the values of a, b, and c if A is orthogonal, where

$$A = \frac{1}{9}\begin{bmatrix} -8 & 4 & a \\ 1 & 4 & b \\ 4 & 7 & c \end{bmatrix}.$$

Ans.: $(a, b, c) = (-1, 8, -4)$ or $(1, -8, 4)$.

(3) Determine the values of a, b, and c when A is orthogonal, where

$$A = \begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix}.$$

Ans.: $a = \pm\frac{1}{\sqrt{2}}, \quad b = \pm\frac{1}{\sqrt{6}}, \quad c = \pm\frac{1}{\sqrt{3}}$, with signs satisfying the orthogonality conditions.

(4)

$$A = \frac{1}{9}\begin{bmatrix} -8 & 4 & 1 \\ 1 & 4 & -8 \\ 4 & 7 & 4 \end{bmatrix}.$$

Ans.: Orthogonal; $A^{-1} = A^T$.

(5) Show that

$$A = \begin{bmatrix} \cos\phi\cos\theta & \sin\phi & \cos\phi\sin\theta \\ -\sin\phi\cos\theta & \cos\phi & -\sin\phi\sin\theta \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

is an orthogonal matrix. **Ans.:** $A^T A = I$.

Q4. Solve the following problems on linear transformations.

(1) Given the transformation

$$Y = \begin{bmatrix} 4 & -5 & 1 \\ 3 & 1 & -2 \\ 1 & 4 & 1 \end{bmatrix} X,$$

find the coordinates (x_1, x_2, x_3) corresponding to $(2, 9, 5)$ in Y. **Ans.:** $x_1 = 2,\ x_2 = 1,\ x_3 = -1$.

(2) Express each of the transformations

$$x_1 = 3y_1 + 5y_2, \qquad x_2 = -y_1 + 7y_2$$

and

$$y_1 = z_1 + 3z_2, \qquad y_2 = 4z_1$$

in matrix form and find the composite transformation expressing x_1, x_2 in terms of z_1, z_2.
Ans.: $X = \begin{bmatrix} 23 & 9 \\ 27 & -3 \end{bmatrix} Z$.

(3) A transformation from the variables x_1, x_2, x_3 to y_1, y_2, y_3 is given by

$$X = AY, \qquad A = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix},$$

and another transformation is given by

$$Y = BZ, \qquad B = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}.$$

Obtain the transformation from x_1, x_2, x_3 to z_1, z_2, z_3. **Ans.:** $X = \begin{bmatrix} 0 & 1 & -1 \\ 3 & 1 & 2 \\ 9 & 7 & 2 \end{bmatrix} Z$.

Q5. Solve the following problems using the Cayley–Hamilton theorem.

(1) Using the Cayley–Hamilton theorem, find A^{-1} and A^5 where

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 0 & 2 \\ 4 & -2 & 1 \end{bmatrix}.$$

Ans.: $A^{-1} = \frac{1}{5}\begin{bmatrix} -4 & 5 & 2 \\ -7 & 10 & 1 \\ 2 & 0 & -1 \end{bmatrix}, \quad A^5 = \begin{bmatrix} 820 & -471 & 529 \\ 575 & -330 & 370 \\ 740 & -422 & 453 \end{bmatrix}.$

(2) Using the Cayley–Hamilton theorem, find A^{-1} and A^4 where

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix},$$

. **Ans.:** $A^{-1} = \frac{1}{4}\begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}, \quad A^4 = \begin{bmatrix} 86 & -85 & 85 \\ -85 & 86 & -85 \\ 85 & -85 & 86 \end{bmatrix}.$

(3) Use the Cayley–Hamilton theorem to find

$$2A^5 - 3A^4 + A^2 - 4I, \qquad A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}.$$

Ans.: $138A - 403I = \begin{bmatrix} 11 & 138 \\ -138 & -127 \end{bmatrix}$.

(4) If

$$A = \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix},$$

find

$$A^4 - 9A^2 + I.$$

Ans.: $2A - 5I = \begin{bmatrix} -3 & 8 \\ 2 & -3 \end{bmatrix}$.

Q6. Solve the following problems on Applications of matrices.

(1) Coordinates of a point P are $(50, 50, 50)$. Origin is shifted to the point $(5, -2, 3)$ with axes transferred parallel to themselves. Rotation is carried out about the z-axis through $45°$. Find the coordinates of the point P in the new coordinate system. **Ans.:** (65.7, 2, 47).

(2) Centre of the arc of the circle in a given coordinate system is $(100, 100, 100)$. Origin is shifted to the point $(-10, -5, 2)$. Rotation is carried out about the y-axis through an angle of $30°$. Find the centre of the arc of the circle in the new coordinate system. **Ans.:** (46.66, 105, 134.66).

(3) The centre of the arc of the circle is $(10, 10, 10)$ in a given co-ordinate system. Origin is $(0, 0, 0)$ and then rotation is carried out about the x-axis through $60°$. Find the co-ordinates of the centre in the new co-ordinate system. **Ans.:**
(10, 5, −8.66).

(4) Centre of the arc of the circle is $(10, 10, 10)$. Origin is shifted to the point $(2, 0, 0)$ along with co-ordinate axes and then rotation is carried out about x-axis through $60°$. Find co-ordinates of the centre in new co-ordinate system. **Ans.:** (8, 5, −8.66).

(5) Centre of arc of the circle in a given co-ordinate system is $(100, 100, 100)$. Origin is shifted to the point $(-10, -5, 2)$. Rotation is carried out about y-axis through an angle of $30°$. Find the centre in new co-ordinate system. **Ans.:**
(46.66, 105, 134.76).